Green Scores: Using Machine Learning Techniques to Predict Hourly Renewable Energy Potential for Global Cities

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Introduction

- The problem:
  - Tackle climate change through renewable energy infrastructure
  - Reduce greenhouse gases
  - Move to cleaner energy sources
  - *Where* and *how*
- The motivation:
  - Preserve ecosystems
  - Protect human health
  - Minimize frequency and intensity of natural disasters

Introduction

- Objectives:
  - Create a metric (“green score”) for renewable energy favorability.
  - Use hourly climate data for ten global cities for comparative analysis.
  - Visualize data and analyze patterns and trends.
  - Train a machine learning model using green score as the target label.
  - Apply trained model to climate data from Charleston, SC - is it suitable for green infrastructure?

Data Collection & Preprocessing

- Data description:
  - Open-Meteo
  - Features:
    i. Wind speed, shortwave radiation, temperature, relative humidity, cloud cover, latitude, longitude, city.
  - Relevance:
    i. Wind speed and shortwave radiation makes up green score.
    ii. Temperature, relative humidity, cloud cover affects how much renewable energy actually gets produced.
    iii. Location is important for renewable energy source access.

Data Collection & Preprocessing (Cont.)

- Data collection:
  - Kaggle dataset recreation:
    i. Global Green Energy Pulse: Real-Time Renewable (Credit: Nudrat Abbas).
    ii. Adjusting for new weather variables and length of time– one year of hourly data.

```python
import kagglehub

# Download latest version
path = kagglehub.dataset_download("nudratabbas/global-green-energy-pulse-real-time-renewable")

print("Path to dataset files:", path)
```

```
Downloading to /root/.cache/kagglehub/datasets/nudratabbas/global-green-energy-pulse-real-time-renewable/1.archive...
100%|██████████| 18.8k/18.8k [00:00<00:00, 28.3MB/s]Extracting files...
Path to dataset files: /root/.cache/kagglehub/datasets/nudratabbas/global-green-energy-pulse-real-time-renewable/versions/1
```

```python
globalgreen = pd.read_csv(path + "/global_green_energy_pulse_20260112.csv")
```

```python
api = "https://archive-api.open-meteo.com/v1/archive?latitude=40.71,35.68,51.5,19.07,-33.86,52.52,-23.55,30.04,6.52,39.9&longitude=-74,139.76,-0.12,72.87,
```

```python
data = pd.read_json(api)
```

Data Collection & Preprocessing (Cont.)

- Data preprocessing:
  - No missing data.
  - Aggregation of “city” column.
  - Formula for green score.
  - Time → datetime and aggregated columns for year, month, day, and hour.

```
df.loc[:, "green_score"] = ((0.5 * ((df["shortwave_radiation"] - solar_min) / (solar_max - solar_min)))
+ (0.5 * ((df["wind_speed_100m"] - wind_min) / (wind_max - wind_min))))

#normalized_value = (x - min) / (max - min)
#green_score = 0.5 * solar_normalized + 0.5 * wind_normalized
```

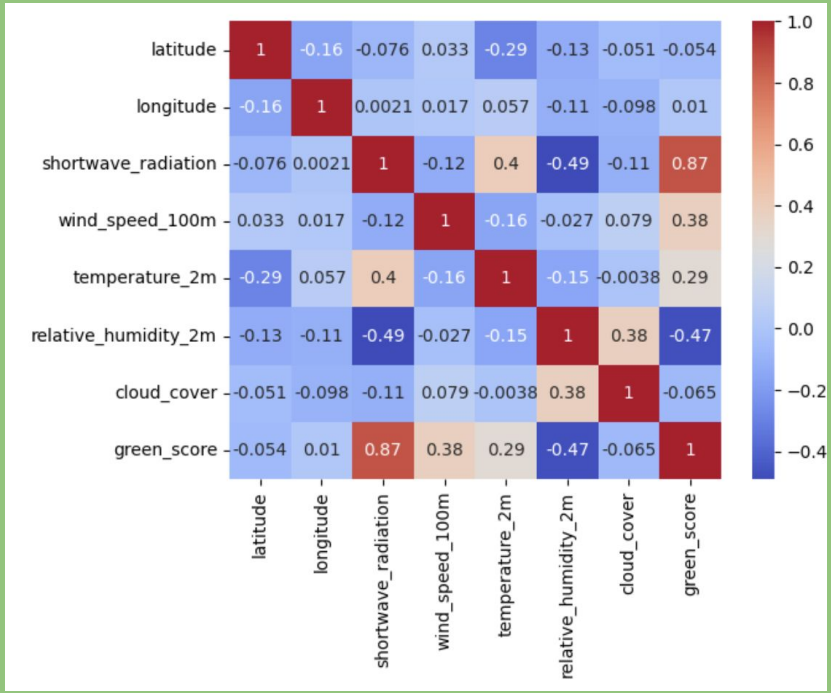
```
df["time"] = pd.to_datetime(df["time"])
df["year"] = df["time"].dt.year
df["month"] = df["time"].dt.month
df["day"] = df["time"].dt.day
df["hour"] = df["time"].dt.hour

df.drop("time", axis=1, inplace=True)
```

```
df.head()
```

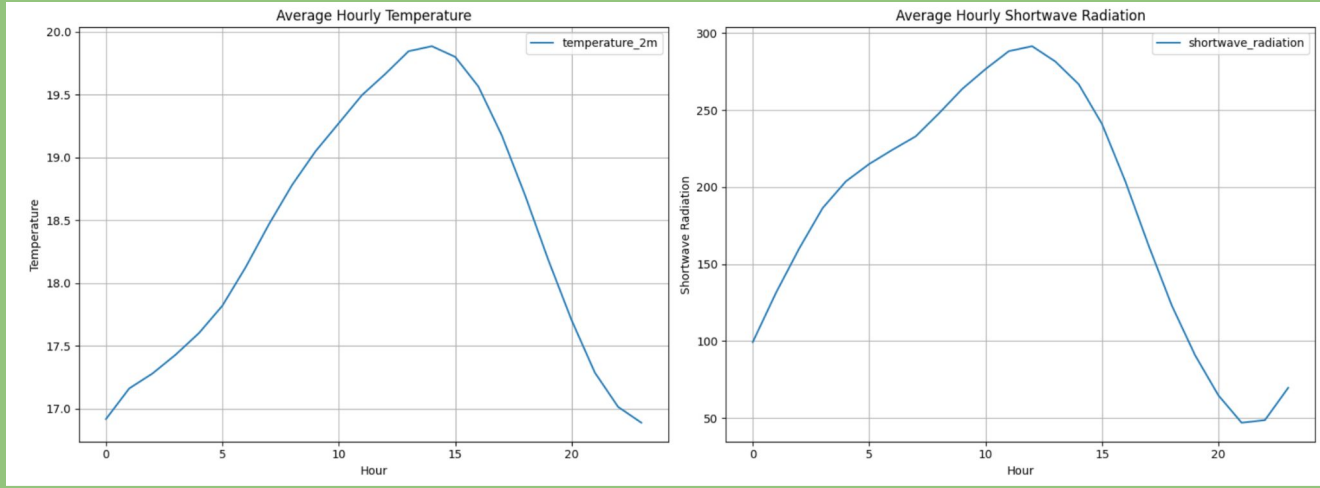
	city	latitude	longitude	shortwave_radiation	wind_speed_100m	temperature_2m	relative_humidity_2m	cloud_cover	green_score	year	month	day	hour
0	New York	40	-74	0.0	24.6	8.2	81	98	0.185801	2025	1	1	0
1	New York	40	-74	0.0	24.3	8.1	82	100	0.183535	2025	1	1	1
2	New York	40	-74	0.0	23.1	7.4	87	100	0.174471	2025	1	1	2

Data Visualization

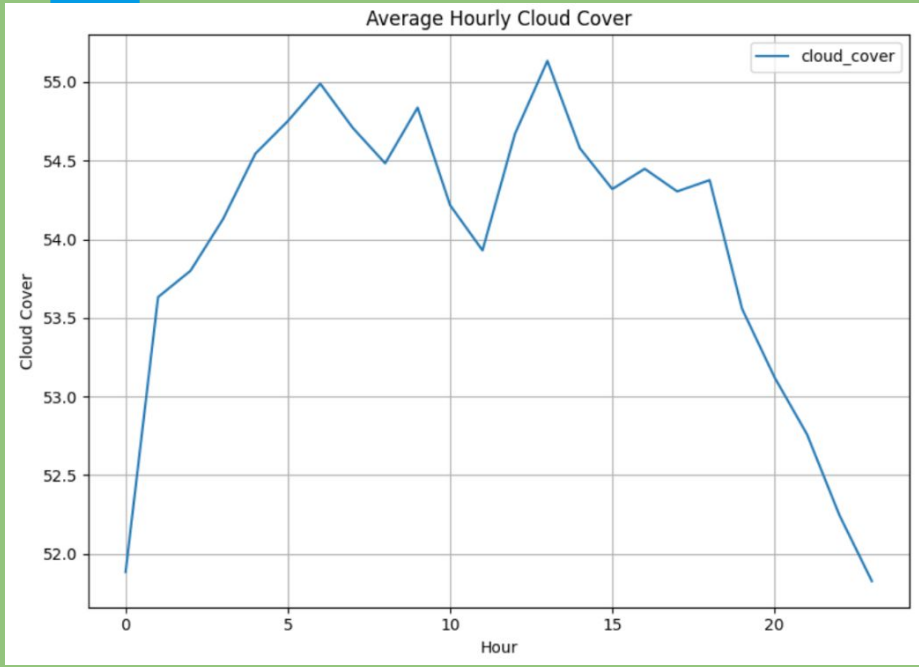
- Goal: explore data and assess which features are important for machine learning step.

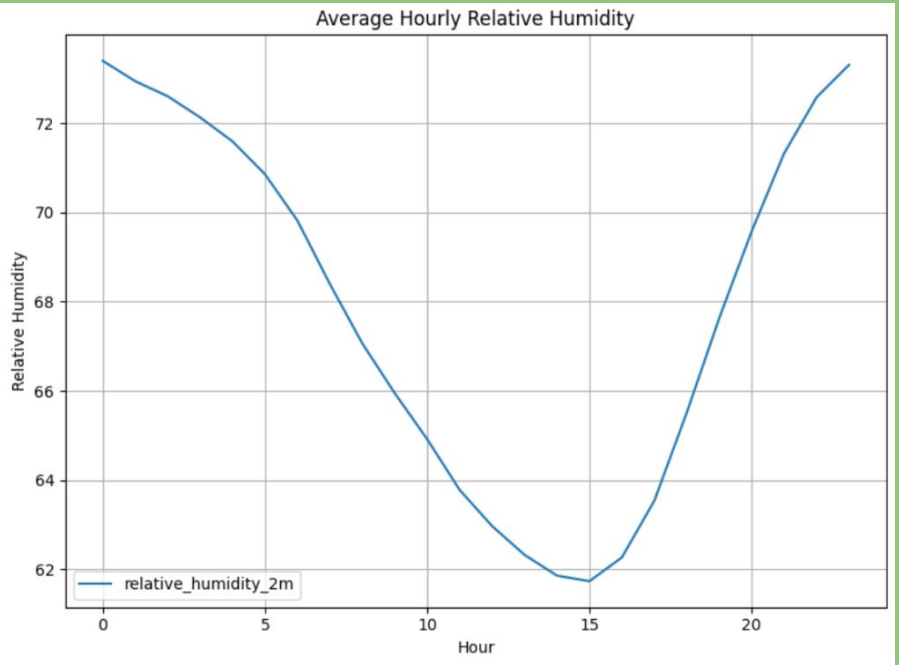


Data Visualization

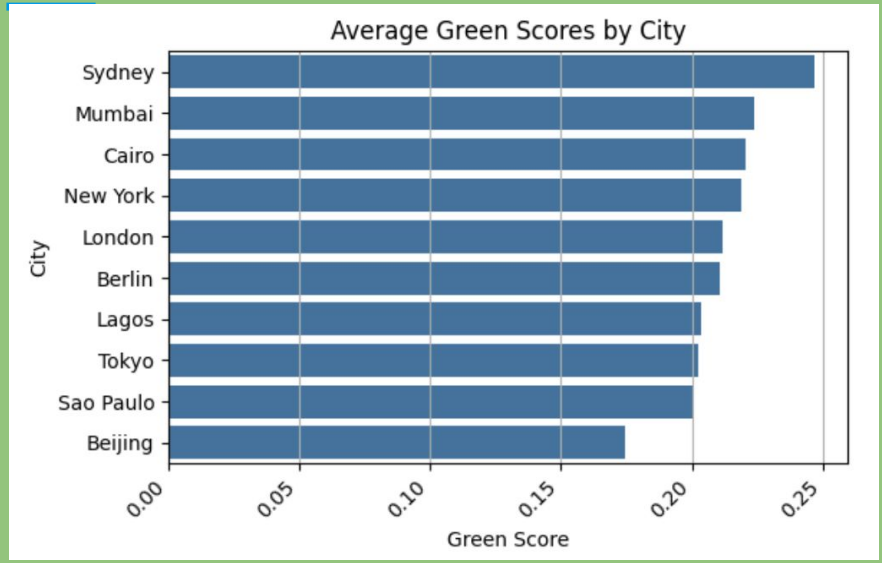


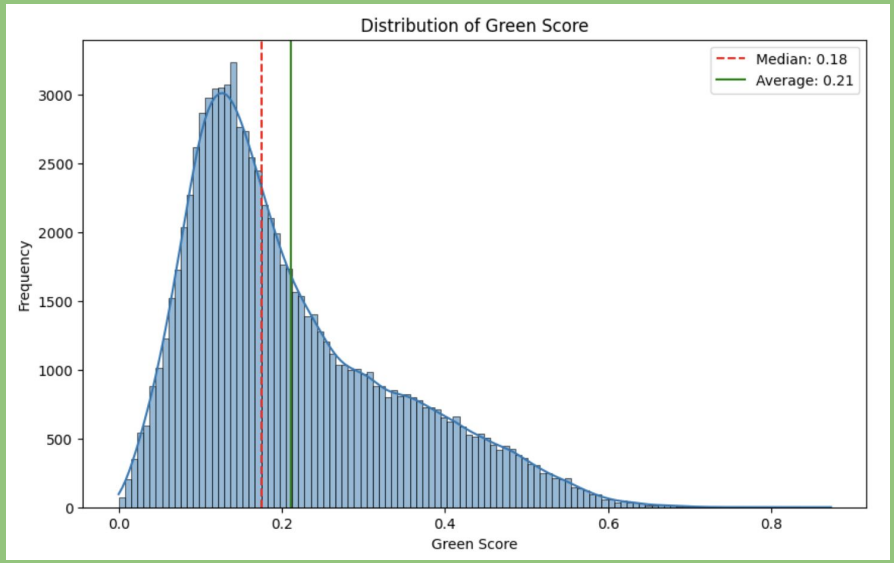
Data Visualization





Data Visualization



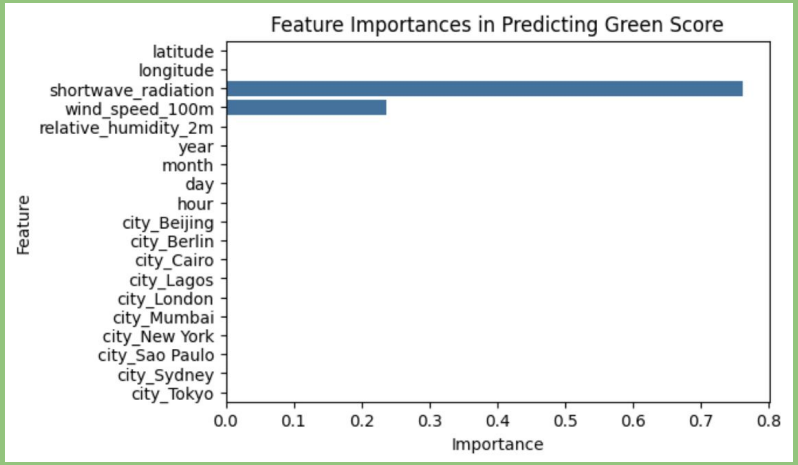


Data Analysis

- Methodology:
  - Continuous target variable.
    i. RandomForestRegressor - advantages.
  - Train model without dropping columns.
  - Train again after dropping columns.

Data Analysis

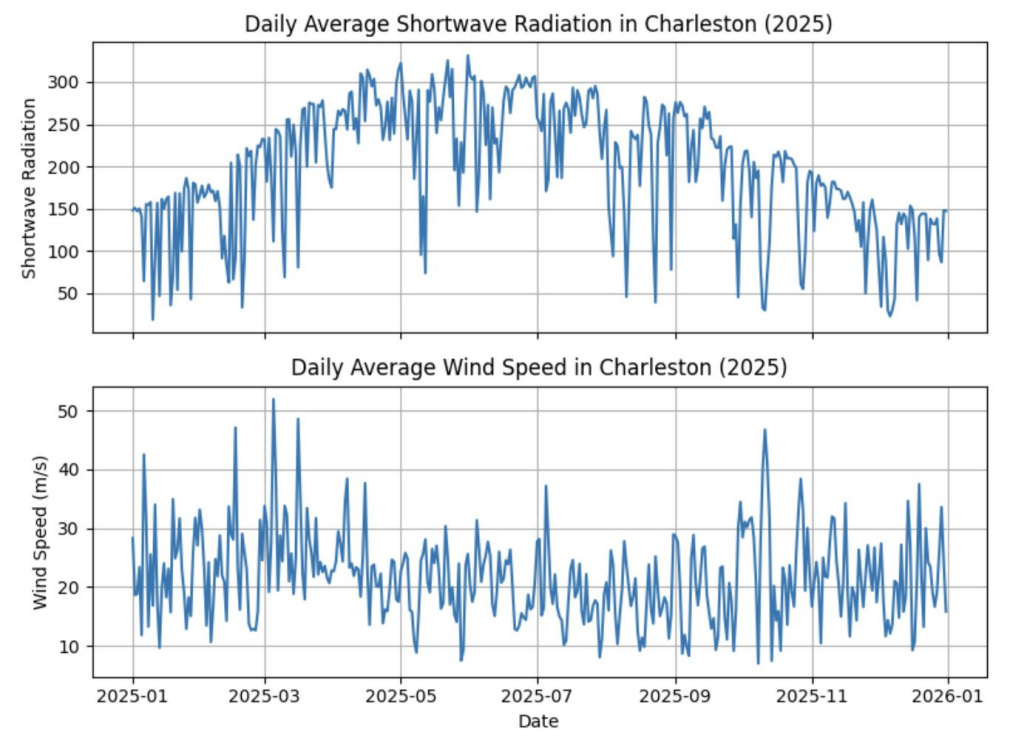
- Results and Interpretation:
  - No change in error.
  - Feature Importance.
  - Cross-validation.




```
Mean Absolute Error (MAE): 0.0004
Mean Squared Error (MSE): 0.0000
Root Mean Squared Error (RMSE): 0.0012
R-squared (R2): 0.9999
```

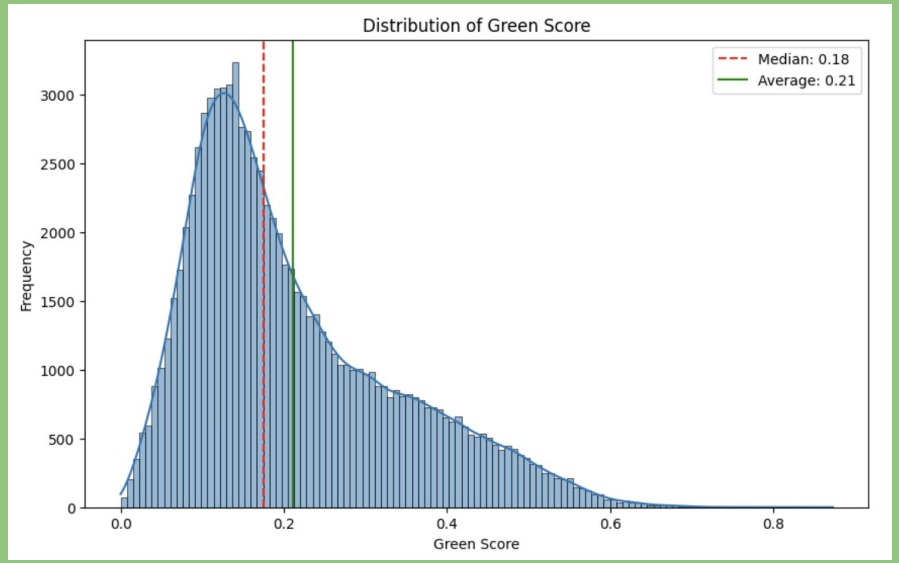
```
Cross-validation R^2 scores: [0.99991234 0.99992249 0.99994527 0.99991583 0.99992241]
Mean cross-validation R^2: 0.9999
Standard deviation of cross-validation R^2: 0.0000
```

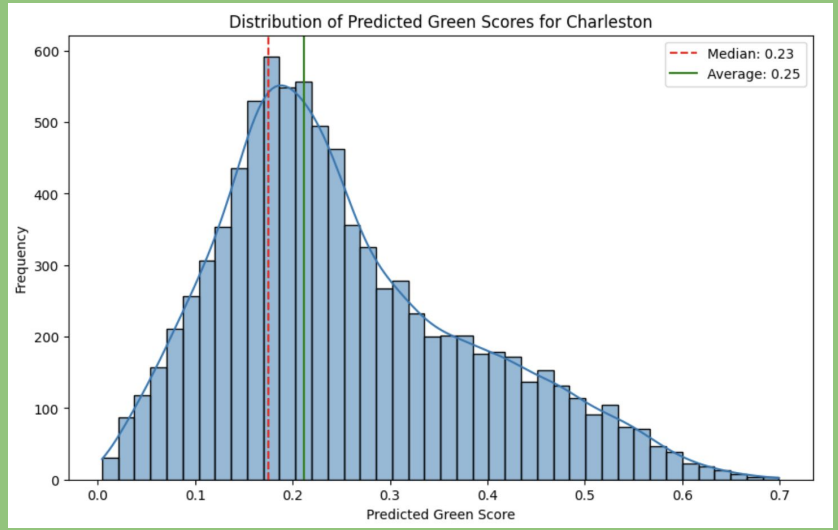
Model Application



Model Application

- Goal: apply trained model to data from Charleston, SC.
  - Perform another API call to Open-Meteo.
  - Hourly time-series data from all of 2025.
  - Same process for cleaning/preparation.





Conclusion

- Successfully engineered “green score.”
- Dataset expansion: patterns and trends.
- Trained a strong RandomForestRegressor model.
- Application to Charleston: predicted green scores.

Future and Related Work

- Similar model with an official sustainability metric.
- Other machine learning algorithms.

- Balasubbareddy Mallala, Azka Ihtesham Uddin Ahmed, Sastry V Pamidi, Md Omar Faruque, Rajasekhar Reddy M, Forecasting global sustainable energy from renewable sources using random forest algorithm. Results in Engineering, Volume 25, 2025, 103789, ISSN 2590-1230, https://doi.org/10.1016/j.rineng.2024.103789.
- Tang, T. Predicting energy prices and renewable energy adoption through an optimized tree-based learning framework with explainable artificial intelligence. *Sci Rep* 16, 6771 (2026). https://doi.org/10.1038/s41598-026-35706-z
- Song Li, Lei Luo, Jun Li, Advanced machine learning approaches for predicting energy and fossil fuel consumption for green growth, Unconventional Resources, Volume 13, 2026, 100262, ISSN 2666-5190, https://doi.org/10.1016/j.uncres.2025.100262.